NARATHIP RODWARNA

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SKILLS

Programming Languages: Python, C++, Java, SQL
Robotics & Simulation: ROS / ROS2 (Gazebo, Rviz, MoveIt), xArm, UR3 Robot Arm, CARLA
Machine Learning & Computer Vision: PyTorch, TensorFlow, ONNX, TensorRT, OpenCV
Other Technical Skills: Docker, Git, Linux, Shell Scripting, CAD, Jira, CAN Bus Protocol, TCP/IP, OSHA-30 Certified

PROFESSIONAL & RESEARCH EXPERIENCE

August Robotics Nevada, USA
Robotics Field Engineer July 2025 – Present

- Deployed and debugged the "Dale" autonomous drilling robot across **19 hyperscale data centers**, diagnosing hardware and software issues on site and remotely to sustain **99.97% drilling accuracy** across 70,000+ holes.
- Built a Python tool for SVD-based rigid registration that maps surveyed reference points to CAD floor plans, replacing manual alignment and cutting site setup time by **30%+**.
- Coordinated site operations, contractors and R&D, turning field edge cases into reproducible logs and data that drove fixes in the autonomy stack.

University of Illinois, Distributed Autonomous Systems Laboratory (DASLAB) Illinois, USA
Graduate Researcher March 2024 – July 2025

- Integrated vision-based crop-row navigation, pest detection and visual servoing on an agricultural robot (TerraSentia base, xArm Lite 6, NVIDIA Jetson Orin) for an NSF-USDA funded precision pesticide-spraying project.
- Developed ROS/MoveIt visual servoing with PI centering, bounding-box depth estimation and IoU-based tracking, aligning the sprayer to pests with a **95% detection rate** and **8.5 s average alignment**.
- Fine-tuned YOLOv8 and benchmarked it against UP-DETR on Jetson Orin for accuracy, speed and power.

Bangkok Mass Transit System Bangkok, Thailand
Rolling Stock Planning Engineer January 2020 – September 2021

- Developed standardized SOPs for maintenance teams, ensuring seamless operational transitions and maintaining high reliability of a driverless Automated People Mover (APM) fleet with **fewer than 1 delay per month**.

Toyota Daihatsu Engineering & Manufacturing Samut Prakan, Thailand
Engineer → Senior Engineer June 2015 – June 2019

- Developed NVH (acoustic) improvement packages for multiple vehicle programs with design teams, and performed component- and vehicle-level validation and verification across lab, test-track and on-road testing.
- Cut acoustic material costs by **~$1.9M per year** through value engineering, optimizing acoustic performance and qualifying cost-effective materials.

PROJECTS

Autonomous Vehicles Perception & Localization February 2024 – February 2025

- Fused IMU, LiDAR and GNSS data with an Extended Kalman Filter (EKF) for vehicle state estimation.
- Estimated vehicle motion with visual odometry using ORB feature matching and essential-matrix decomposition.
- Implemented particle-filter localization with LiDAR measurements for precise indoor localization.
- Calibrated a Velodyne LiDAR and ZED stereo camera (intrinsics and extrinsics via ICP point-cloud registration), and deployed and validated the perception stack on a physical autonomous test vehicle.

Visual SLAM Benchmarking & Failure Analysis (sponsored by GE Aerospace) May 2024 – November 2024

- Benchmarked Visual SLAM algorithms (ORB-SLAM3, OpenVSLAM, LSD-SLAM) on the EuRoC and Kagaru airborne datasets, quantifying localization accuracy with APE/RPE and comparing recovery behavior after tracking loss.
- Built a time-series failure-analysis pipeline (ADF stationarity tests, Granger-causality feature selection, AIC lag search, VAR/AutoReg) on image and motion features to test early-warning signals for SLAM tracking loss.

EDUCATION

University of Illinois Urbana-Champaign Illinois, USA
Master of Engineering in Autonomy and Robotics

Chulalongkorn University Bangkok, Thailand
Bachelor of Engineering in Mechanical Engineering